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Overview

In this phase of the project, the role of Fleet Manager for Az National Trucking (ANT), a fictitious national trucking corporation based in California, is assumed. The primary responsibility entails ensuring compliance among all fleet drivers with the corporation's regulations, thereby mitigating insurance risks associated with factors such as speeding, unsafe following, lane departure, and other hazardous driving practices. The pertinent details regarding the company's profile and contextual scenario are outlined below.

Company Overview

Established a decade ago, ANT comprises 400 employees, predominantly long-distance truck drivers. The core operation of ANT involves offering long-haul trucking services for various general and non-specialized cargo types, excluding hazardous materials (HAZMAT), across the western United States. ANT holds licensure and approval for transportation services in 14 states. The organizational structure and essential details are elaborated as follows:

Organizational Structure: ANT operates as a privately held entity, characterized by the following breakdown:

- Total Employees: 400
  - Staff: 40
  - Mechanics: 15
  - Administration: 20
  - Management: 15
  - Drivers: 310
    - W2 Employees: 300
    - 1099 Contractors: 10

Trucking Licensure Approval:

- States: AZ, CA, CO, NV, UT, WA, OR, ND, SD, KS, MO, NE, NM, WY
- Countries and Provinces:
  - Canada: BC, Alberta, Saskatchewan
  - Mexico: Baja only

Hub Locations:

- Headquarters: City of Industry, California
- Secondary Hubs:
  - Salt Lake, UT
  - Seattle, WA
  - Fargo, ND
  - Kansas City, MO

Fleet Composition:

- Trucks Owned by ANT: 300
- Individually Contracted: 10

Compliance Guidelines: ANT adheres to compliance and risk management practices as dictated by the Federal Motor Carrier Safety Administration (FMCSA) regulations for the trucking industry, meeting or surpassing the minimum requirements specified in relevant sections. Additionally, compliance extends to the Department of Transportation regulations and pertinent state-specific laws and regulations. ANT retains the prerogative to impose more stringent rules concerning certain risk factors, as warranted by telematics data or business exigencies, within the bounds of legal permissibility.

Risk Thresholds

Risk mitigation thresholds are established based on regulatory mandates, business imperatives, and telematics data observations. The organization's current risk factors encompass various parameters, including but not limited to:

- Mileage
- Speed
- State highway law compliance
- Number of trailers towed
- Freight weight (tonnage)
- Driver logs
- Cargo limitations
- Off-hours operations
- Drug test results
- Driving records
- Telematics sensor violations
- Road closures and rerouting
- Vehicle maintenance and inspection
- Driver qualifications

Truck Driver Risk Factor Cases:

In this phase, the "Mileage/Speed" risk factor threshold is employed to assess individual drivers' risk levels. A risk factor exceeding or equaling 7.0 (on a scale of 10) triggers an alert or system notification to ANT corporate management and insurance providers. The risk factor for each driver is calculated by dividing the total miles by the risk factor, as detailed in the provided Geolocation Excel file.

Task Description:

1. Transfer the remaining input file into HDFS.
2. Follow the instructions provided in the "DDLRISK_FACTOR_UTD.DOCX" document to create and load the data.
3. Locate and validate all tables from Tableau.
4. Create the appropriate schema model in Tableau.
5. Discuss with your group to generate 5 different business questions. Each question should be answered through a chart in Tableau.
6. Create a Dashboard that displays all charts collectively.
7. Be prepared to demonstrate each business question and its corresponding answer from the Dashboard.

Visualizations and Reports Samples

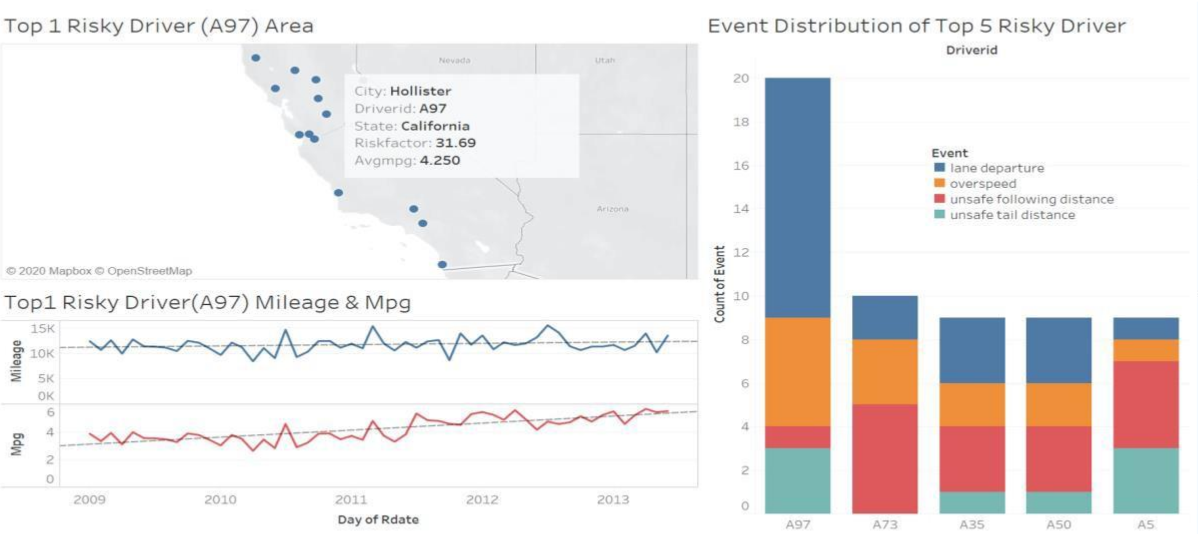
From a couple of visualizations, the following two sample reports can be concluded:

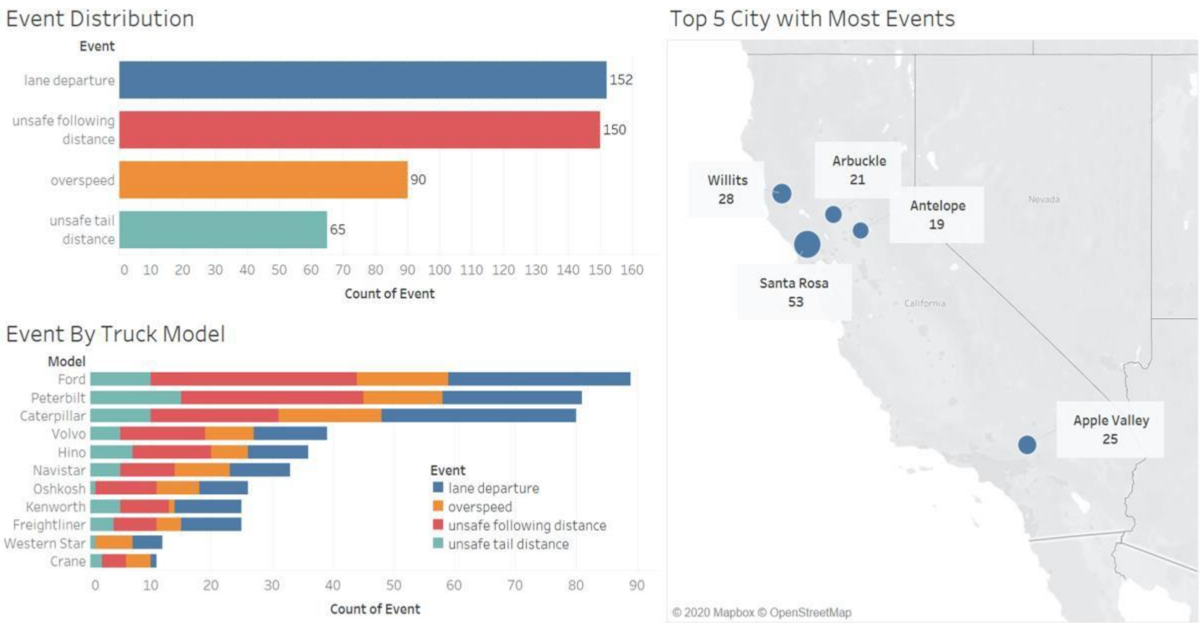
Report#1 from one of the business questions:

Truck id # A88 – (Hino model) drove a total of 9,653 miles in the city of San Diego in the month of April of 2015 with an average of 5.4 mpg and a risk factor of 7.51 (on a scale of 1-10).

Report#1 from one of the business questions:

Truck id # A40 – (Ford model) drove a total of 10,876 miles in the city of Holtville in the month of April, 2015 with an average of 5.8 mpg and a risk factor of 3.1 (on a scale of 1-10).





Deliverables:

- Collaborate with team members to analyze provided data and generate charts highlighting drivers' risk factors.
- Develop a concise PowerPoint presentation showcasing the analysis results, limited to 8 slides, covering:
    1) Problem statements and objectives
    2) The 5 business questions
    3) Workflow diagram illustrating the analysis process, including utilized components and ecosystems.
    4) Dashboard or report chart displaying risk factors per driver and highlighting those surpassing the threshold.